

SQL Assignment

Topics Covered

1. Database and Table Creation
2. INSERT, UPDATE and DELETE Queries
3. SELECT, WHERE and Sorting
4. Aggregate Functions
5. GROUP BY and HAVING
6. Joins
7. Constraints
8. Mini Database Design

Total Questions: 20

Section A: Database and Table Creation

Question 1: Create a Student Database

Create a database named **CollegeDB**. Inside it, create a table named **Students** with the following columns:

- Student_ID
- Student_Name
- Age
- Gender
- Course
- City

Choose appropriate data types and set **Student_ID** as the Primary Key.

Question 2: Create an Employee Table

Create a table named **Employees** with the following columns:

- Employee_ID
- Employee_Name

- Department
- Salary
- Joining_Date
- Email

Ensure that:

- Employee_ID is the Primary Key.
 - Email must be unique.
 - Salary cannot be negative.
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Question 3: Create a Product Table

Create a table named **Products** containing:

- Product_ID
- Product_Name
- Category
- Price
- Stock

Use appropriate constraints for all columns.

Question 4: Create a Course Table

Create a table named **Courses** with:

- Course_ID
- Course_Name
- Duration
- Fees

Set suitable constraints for each column.

Question 5: Create a Customer Table

Design a table named **Customers** that stores:

- Customer_ID
- Name
- Phone_Number

- Email
- City

Apply appropriate constraints wherever necessary.

Section B: INSERT, UPDATE and DELETE

Question 6: Insert Student Records

Insert at least **10 student records** into the **Students** table with different courses and cities.

Question 7: Update Employee Details

Perform the following operations:

- Increase the salary of employees working in the IT department by 10%.
 - Change the department of one employee.
 - Update the email address of a selected employee.
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Question 8: Delete Records

Delete:

- Students whose age is less than 18.
 - Products with zero stock.
 - Customers belonging to a specified city.
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Question 9: Insert Product Records

Insert **10 products** into the **Products** table with different categories and prices.

Question 10: Customer Record Maintenance

Insert five customer records, then:

- Update one customer's city.

- Update another customer's phone number.
 - Delete one customer record.
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Section C: SELECT and Filtering

Question 11: Retrieve Student Information

Write SQL queries to:

- Display all students.
 - Display only student names and courses.
 - Display students belonging to a particular city.
 - Display students older than 20 years.
 - Sort students by name in ascending order.
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Question 12: Employee Search

Write queries to:

- Display employees earning more than ₹50,000.
 - Display employees from a particular department.
 - Display employees whose names start with the letter 'A'.
 - Display employees ordered by salary in descending order.
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Question 13: Product Queries

Write SQL queries to:

- Display products costing more than ₹1000.
 - Display products between ₹500 and ₹2000.
 - Display products belonging to a particular category.
 - Display the five most expensive products.
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Question 14: Customer Queries

Write SQL queries to:

- Display customers from a specified city.
 - Display customers whose names end with "Kumar".
 - Display customers ordered alphabetically.
 - Display unique city names.
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Question 15: Practice Using SQL Operators

Using any table created above, write queries using:

- AND
 - OR
 - NOT
 - IN
 - BETWEEN
 - LIKE
 - ORDER BY
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Section D: Aggregate Functions and GROUP BY

Question 16: Employee Salary Analysis

Write SQL queries to find:

- Total salary of all employees.
 - Average salary.
 - Highest salary.
 - Lowest salary.
 - Total number of employees.
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Question 17: Product Statistics

Write queries to calculate:

- Average product price.
- Total stock available.
- Highest priced product.
- Lowest priced product.

Question 18: GROUP BY and HAVING

Using the Employees table:

- Count employees department-wise.
 - Calculate average salary department-wise.
 - Display only departments having more than three employees.
 - Display departments whose average salary is greater than ₹60,000.
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Section E: Joins and Mini Project

Question 19: SQL Joins

Create two tables:

- **Students**
- **Courses**

Each student should be enrolled in one course.

Write SQL queries using:

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN

Display student names along with their course names.

Question 20: Mini Database Project

Design a simple **Library Management System** database.

Create the following tables:

- Books
- Members
- Borrow_Records

Perform the following tasks:

- Insert at least five records into each table.
- Retrieve all borrowed books with member names using JOIN.
- Display books that are currently available.
- Count the total number of books borrowed by each member.
- Display the member who has borrowed the maximum number of books.